

Adsorption enhancement of Congo red dye from wastewater based on edamame shell originated activated carbon by the cations: Experimental and theoretical studies

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ARTICLE INFO

Keywords:

Edamame shell
Activated carbon
Congo red
Adsorption
Cations

ABSTRACT

In this study, a microporous activated carbon (AC) with excellent performance was successfully prepared by KOH impregnation activation method using edamame shell as raw material. The synthesized AC with high specific surface area and pore volume can be effectively used for the adsorption and removal of Congo red (CR) from wastewater. Nitrogen adsorption isotherms, Scanning electron microscopy (SEM), X-ray diffraction (XRD), Fourier transform infrared (FTIR), X-ray photoelectron spectroscopy (XPS) and zeta potential were used to characterize the structure of activated carbon. The ultra-high surface area and well-developed pore volume provided sufficient adsorption sites for CR adsorption, and the maximum adsorption effect of AC on CR was up to 1509.60 mg/g at 25 °C. The effects of pH, contact time, temperature and adsorbent mass on CR adsorption were investigated. In addition, the effects of different cations (Ca^{2+} , Na^+ , K^+ , NH_4^+) on the CR adsorption of anionic dyes were focused by adding different concentrations (10 mM, 20 mM, 30 mM) of cations. The results showed that Ca^{2+} , Na^+ and K^+ had a significant promotion effect on the CR adsorption of AC, among which Ca^{2+} had the most obvious promotion effect. Through model fitting, CR adsorption by AC conforms to Liu adsorption isotherm model and Elovich kinetic model. The results showed that the addition of cations could substantially improve the adsorption capacity of AC on the anionic dye CR, which has potential application in the field of dye wastewater treatment.

1. Introduction

Water pollution is a growing global crisis. Organic dyes are widely used in industrial production, which have a great impact on the environment and have become one of the largest sources of water pollution. Moreover, most organic dyes are carcinogenic, hard to degrade and highly toxic [1–3]. According to reports, nearly 2 million tons of dye-containing industrial effluents are discharged into the world's water resources every day [4]. Studies have shown that low concentrations of dyes in water are harmful to terrestrial and aquatic organisms [5]. Congo red ($\text{C}_{32}\text{H}_{22}\text{N}_6\text{Na}_2\text{O}_6\text{S}_2$) is benzidine anionic azo dyes with double azo functional groups containing long-chain linear molecule and two symmetrical naphthylamine groups [6]. It shows good solubility and poor biodegradability when goes into the wastewater treatment system. Through the metabolic process, CR can enter the body to the skin, sight, smell, harmful threat posed by the nervous system and respiratory

system. Therefore, the removal of CR dye from industrial effluents is of paramount importance.

At present, many engineering methods have been studied in order to remove organic dyes from dye wastewater and to minimize the pollution caused by them to the environment. Many advanced methods like adsorption [7], flocculation [8], ion exchange [9], membrane separation [10], and photocatalytic degradation [11] have achieved remarkable results in the treatment of these wastewaters. Among these methods, adsorption is the most common wastewater treatment technology due to its low cost, high removal efficiency and less sludge production [12]. Various adsorbents have been used for adsorption and treatment of dyes in wastewater, and among these materials, porous materials have received much attention due to their properties such as high specific surface area [13]. Wang et al. [14] prepared a graded porous carbon material using egg white as a raw material with an adsorption capacity of 869.73 mg/g for Congo red dye. Metal oxide nanomaterials, due to

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<https://doi.org/10.1016/j.diamond.2023.109930>

Received 19 December 2022; Received in revised form 10 March 2023; Accepted 12 April 2023

Available online 17 April 2023

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their stable physicochemical properties, high electrical conductivity [15], widely used not only in the field of electrode materials, but also in the field of wastewater treatment. Bastami et al. [16] prepared a new nanostructured material $\text{Cu}_2\text{O}/\text{Fe}_2\text{O}_3/\text{MoC}$ by ultrasound-assisted counter-precipitation method. The newly prepared adsorbent material with well-developed pore structure is more favorable for dye adsorption and the maximum adsorption capacity of dye RY195 can reach 434.78 mg/g, which has high potential for application. Amran et al. [17] prepared activated carbon by activating *Casuarina* empty fruit with different ratios of NaOH and the adsorption capacity of Congo red could reach 232 mg/g.

Among many adsorbents, activated carbon has shown satisfactory performance, such as porous structure, diverse functional groups and easy availability of raw materials, and has become the material of choice [18,19]. In addition, carbon-based materials such as activated carbon have also been used to prepare electrode materials for capacitors due to their large specific surface area and stable chemical properties [20,21]. Among the many raw materials for the preparation of activated carbon, agricultural by-products are an excellent choice because of their cheap raw materials and abundant sources. Agricultural by-products are wastes of agricultural products, which are harmful to the environment, and how to deal with these wastes has become a major economic and ecological problem, and using agricultural wastes as activated carbon is a good solution [22,23]. Soybean is an important cash crop, mainly distributed in North America, South America, East Asia and South Asia. In 2017, global and Chinese annual production was 353 million tons and 14 million tons, respectively [24]. Edamame is popular in China, but the large quantities of the shells produced are often difficult to dispose of as agricultural waste which can be harmful to the environment. However, with high cellulose content, edamame shell can be used as biological material for the preparation of activated carbon.

In this study, impregnated with KOH, the activated carbon from edamame shell was used for CR adsorption. It successfully realized the recycling of agricultural waste and reduced its pollution to the environment. The prepared AC has super high specific surface area and pore volume, and the adsorption capacity of Congo red exceeds that of most similar adsorbent materials reported in the literature. In addition, the effect of cations on the adsorption CR of AC was further analyzed in this paper. We focused on further improving the adsorption effect of AC by adding cations, and found that the addition of K^+ and Na^+ could appropriately improve the adsorption effect of AC on CR, and the addition of Ca^{2+} could greatly improve the adsorption effect of AC on CR. The addition of cations to improve the adsorption effect of AC on CR has good potential for application in the environmental protection field.

2. Experimental

2.1. Materials

The experimental edamame shells were purchased from a farmer's market in Wuhan, Hubei Province, China. Congo red was purchased from Shanghai Aladdin Biochemical Technology Co. The chemical reagents used in this study are all of analytical grade and can be used directly in the experiments. All solutions in the experiments were diluted with deionized water.

2.2. Preparation of edamame shell-based activated carbon (AC)

The edamame shells used in the experiment need to be pretreated. Firstly, a certain amount of edamame shells were taken and washed with water to remove surface impurities. Then the edamame shells were washed with 1 M NaOH solution and 1 M HCl solution in turn. Finally, the acid-washed edamame shells were washed several times with deionized water to remove the residual hydrochloric acid on the surface. Until the edamame shells were neutral, and then they were dried in a drying oven at 80 °C for 24 h. The dried edamame shells were put into a

grinder and ground into hull powder and sieved. A certain amount of edamame shell powder was placed in a tube furnace to be carbonized at N_2 flow (100 mL/min) & 500 °C for 120 min. Then the biochar can be obtained.

The obtained biochar was mixed with KOH in a 1:3 (weight ratio) impregnation, placed in a drying oven and dried under 110 °C for 12 h. The impregnation was then placed in a tube furnace. The biochar is then activated under N_2 flow (100 mL/min) & 800 °C for 90 min. Afterwards, the activated samples were removed, acid-washed with 0.1 M HCl solution, and then washed with deionized water to neutrality to obtain edamame shell activated carbon (AC).

2.3. Characterization

The nitrogen adsorption/desorption isotherms of the samples were determined with a Micromeritics ASAP 2020 analyzer for true determination. The surface morphology of the AC samples was analyzed by scanning electron microscopy (SEM), and the surface functional groups of the samples before and after adsorption were investigated by intelligent Fourier transform infrared spectroscopy in the range of 4000–400 cm^{-1} . The AC samples before and after adsorption were analyzed using x-ray photoelectron spectroscopy (XPS) and x-ray diffraction (XRD) maps. The geometry type used for XRD was Bragg Brentano, the scan speed was 5 °C/min, 2θ was scanned from 5° to 80°, and the scan intervals were 0.01°. The zeta potential was analyzed using a nano particle size and zeta potential analyzer (DLS).

2.4. Batch adsorption experiments

The adsorption experiments of Congo red were carried out by intermittent adsorption method. Congo red powder was dissolved into deionized water to prepare Congo red mother liquor (1000 mg/L), and then the 1000 mg/L Congo red solution was diluted to the desired concentration of the experimental solution by using deionized water. Unless otherwise stated, all adsorption experiments were performed by mixing 10 mg of AC and 150 mL of CR solution in a conical flask, the concentration of CR solution is 200 mg/L, and experiments were performed at an ambient temperature of 25 °C. Then it was shaken continuously in a constant temperature shaker (LICHEN ZD-85, China) at a rate of 200 $\text{r}\cdot\text{min}^{-1}$ for 5 h to ensure adequate adsorption. The dye concentration was detected at certain intervals during adsorption, and a small amount of the supernatant was filtered and the concentration of CR in the filtrate was measured at 497 nm using a UV-visible spectrophotometer (METASH UV-5500, China). All adsorption experiments were repeated 3 times to ensure reproducibility of the experiments. The pH of the solution was adjusted by adding a small amount of HCl (0.1 M) or NaOH (0.1 M) solution dropwise to the CR solution. The concentrations of the different cations were adjusted to the experimentally desired values by adding cationic chloride salts (NH_4Cl , NaCl , KCl , CaCl_2). Amount of CR adsorption, q_e ($\text{mg}\cdot\text{g}^{-1}$), and the removal rate, R (%), was determined by:

$$q_e = (C_0 - C_e) \frac{V}{m} \quad (1)$$

$$R = \frac{(C_0 - C_e)}{C_0} \times 100\% \quad (2)$$

where C_0 (mg/L) is the initial concentration of CR in the solution before adsorption, C_e (mg/L) is the concentration of CR in the solution after adsorption, V (L) and m (g) are the volume of CR solution and the mass of adsorbent AC, respectively.

2.5. Adsorption models

All adsorption isotherm models, adsorption kinetic models and thermodynamic equations used in this study are detailed in Table S1.

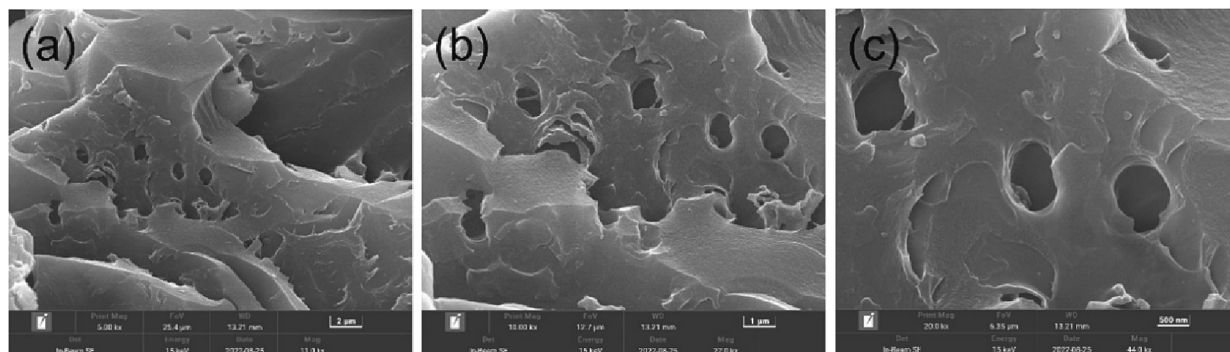


Fig. 1. SEM images of AC: (a) magnification 5000 times; (b) magnification 10,000 times; (c) magnification 20,000 times.

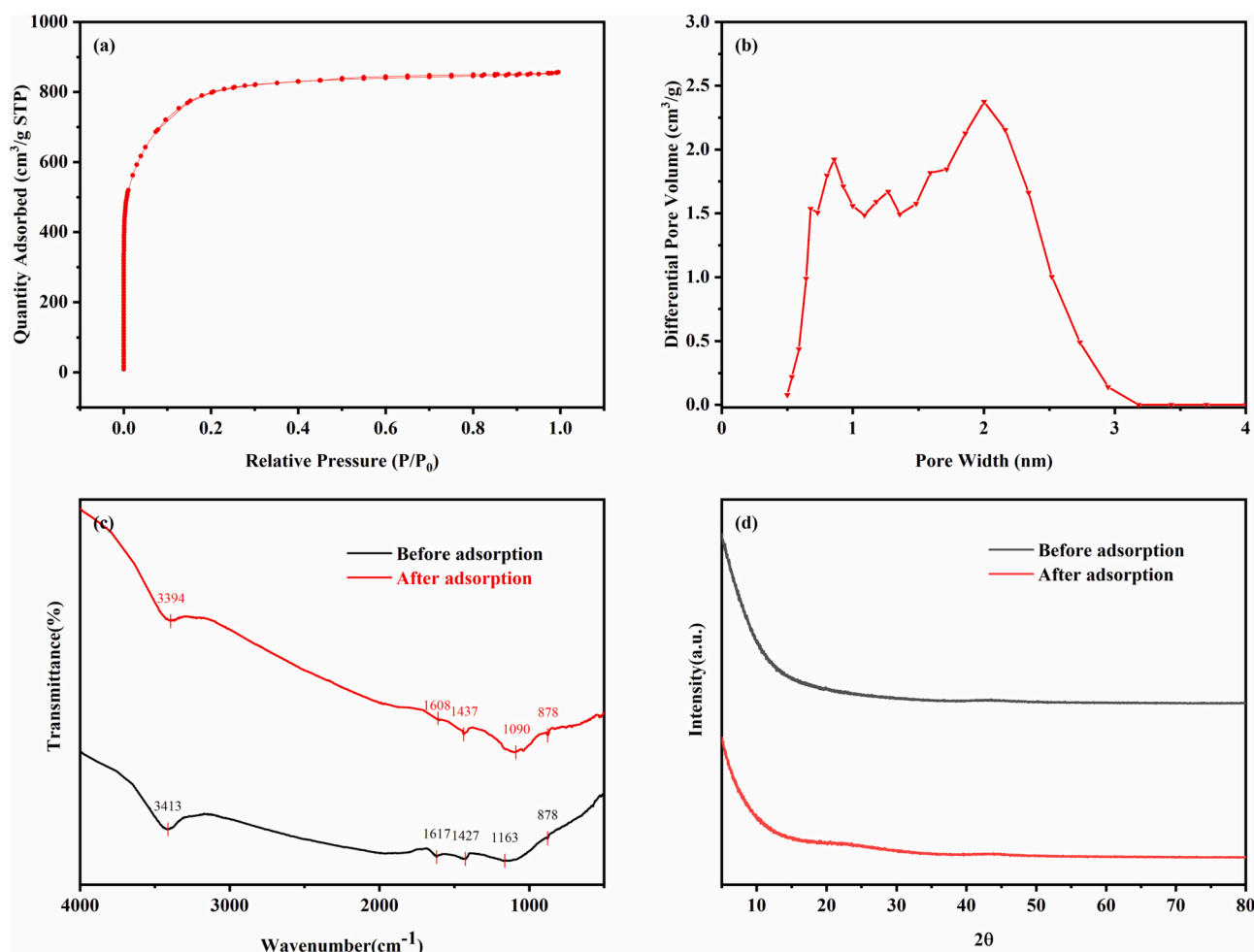


Fig. 2. (a) N_2 adsorption/desorption isotherms of AC; and (b) pore size distribution of AC; and (c) FT-IR spectra of AC before and after adsorption of CR; and (d) XRD patterns of AC before and after adsorption of CR.

3. Results and discussion

3.1. Characterization of AC

Fig. 1 shows the SEM images of AC at different magnifications, and it can be found that the surface of the prepared AC is rough and many pores are formed on the surface. This may be because the KOH impregnated at high temperature is in full contact with the raw material of edamame shells carbon, and with the increase of temperature activation reaction occurs between them, some small molecule gases such as CO_2 and CO are produced. These small molecule gases produce a large

number of pores on the surface. These pore structures can provide conditions for the adsorption of AC.

As illustrated in Fig. 2a, the N_2 adsorption/desorption isotherm of AC is type I, but h4-type hysteresis lines exist in the region of relative pressure 0.50–0.95 P/P_0 , indicating that the material has both microporous and mesoporous structures. This unique pore structure allows AC to obtain an ultra-high specific surface area and total pore volume [25]. Fig. 2b shows the pore size distribution curve of AC, and it can be found that a large number of pore sizes are distributed in the micropore (<2 nm) range, which further proves that the sample has a large number of microporous structures. The pore structure properties of AC were

Table 1

The pore textural properties of AC.

Sample	S_{BET} ($\text{m}^2\cdot\text{g}^{-1}$)	V_{mic} ($\text{cm}^3\cdot\text{g}^{-1}$)	V_{total} ($\text{cm}^3\cdot\text{g}^{-1}$)
AC	2653.382	0.702	1.324

obtained by BET and BJH analysis. As shown in Table 1, the microporous volume of AC (V_{mic}) is $0.702 \text{ cm}^3\cdot\text{g}^{-1}$, the total pore volume (V_{total}) is $1.324 \text{ cm}^3\cdot\text{g}^{-1}$, and the specific surface area (S_{BET}) reached $2653.382 \text{ m}^2\cdot\text{g}^{-1}$. The parameters of AC were higher than those of most activated carbons reported in the literature.

Fig. 2c shows the FTIR spectra of AC before and after adsorption of CR dye. Which can be used to analyze the functional groups on the surface of AC. It can be observed that the spectra of AC before and after adsorption are very similar. After adsorption of the dye, an increase in the intensity of the main bands of the FTIR spectrum can be observed, and these phenomena suggest that the adsorption of CR dye on AC is mainly related to physical interaction forces [26]. The peak that can be observed at 3413 cm^{-1} may be due to the stretching vibration of $-\text{OH}$. After adsorption, the broad and strong $-\text{OH}$ stretching vibrational band of AC at 3413 cm^{-1} shifted to 3394 cm^{-1} , indicating the existence of hydrogen bonding between FR and the dye [27]. The bands at 1617 and 1608 cm^{-1} can be attributed to the stretching vibration of $\text{C}=\text{O}$ [28]. The peaks appearing at 1163 cm^{-1} and 1090 cm^{-1} may be related to the stretching vibration of the $\text{C}-\text{O}$ bond, which may be attributed to the carboxyl group on the surface of the AC. The shift of the stretching

vibration band before and after adsorption is due to the strong adsorption of the Congo red dye by AC. The diffraction patterns of AC before and after the adsorption of CR are shown in Fig. 2d, AC exists in a typical non-crystalline structure, this may be related to the lignin content contained in AC [29]. In addition, the diffractograms of CR adsorption by activated carbon did not change significantly, indicating that the adsorption process did not change the crystallinity of the material.

XPS was used to analyze the surface chemistry of the AC under investigation. The spectra show that AC contain two main elements, with carbon as the major element and oxygen as the minor element (Fig. 3a). In the C 1s spectra (Fig. 3b), the XPS peaks of AC are located at 284.8 eV , 286 eV and 290 eV , which are attributed to $\text{C}-\text{C}$, $\text{C}-\text{O}$ and $\text{C}=\text{O}$ on AC, respectively [30]. Fig. 3c shows the O 1s spectrogram of AC with the XPS peaks of AC at approximately 532 eV and 533.2 eV due to the $\text{C}-\text{O}$ bond and $\text{C}=\text{O}$ bond [31]. Fig. 3d shows that the AC sample before adsorption did not contain S element, but after adsorption the peak of S 2p could be detected. The CR dye molecule was the only substance with S elements present in the experiment, which also indicates that the CR adsorption experiment was successfully performed. These analyses also agree well with the results of the FTIR analysis above.

3.2. Congo red adsorption studies

3.2.1. Effect of solution pH

Solution pH has a significant effect on the adsorption of dyes. At

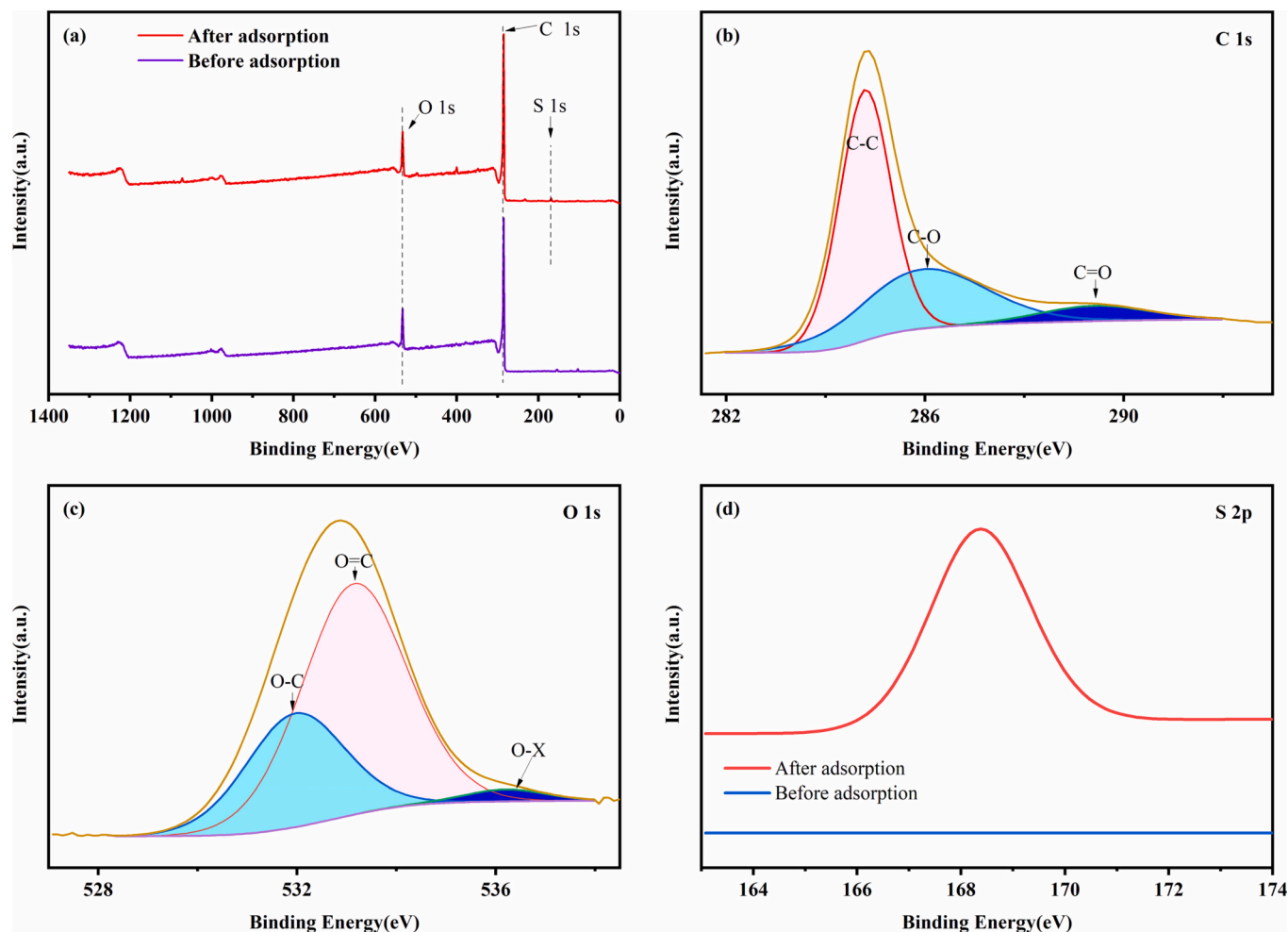


Fig. 3. XPS full spectra of AC before and after adsorption of CR (a); high-resolution spectrum of C 1s (b) and O 1s (c) of AC; high-resolution spectrum of S 2p of AC before and after adsorption of CR(d).

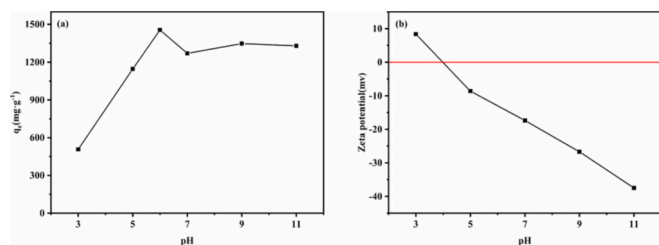


Fig. 4. Effect of pH on CR sorption by AC(a); The pH_{pzc} of AC (b).

different pH, different concentrations of H^+ and OH^- will be present in the solution, and these H^+ and OH^- will compete with the contaminants for the active sites present on the adsorbent. Moreover, pH also affects the surface properties of the AC, such as the protonation and deprotonation of functional groups on the AC. The pH of the solution also leads to changes in the structure of the CR [32,33]. In this study, the effect of different pH values on CR adsorption was investigated by adjusting the pH value of the solution in the range of 3–11. The experiments also determined the zeta potential of AC at different pH.

The effect of the initial pH on the adsorption efficiency of AC was investigated in the pH range of 3–11, as shown in Fig. 4a. The zeta potential analysis and effects were shown in Fig. 4b, indicating that the point of zero charge (pH_{pzc}) of AC is around 4. The zeta potential of AC is very sensitive to the pH of the aqueous solution, which gradually decreases with the increase of pH. When $pH < pH_{pzc}$, the surface of AC is positively charged due to the presence of large amount of H^+ in the solution. When $pH > pH_{pzc}$, the surface of AC is negatively charged due to the presence of large amounts of OH^- . Since the dissociation constant (pK_a) of CR is 4.1, when $pH < 4.1$, CR molecules exist mainly as cations, and when $pH > 4.1$, CR exists mainly as anions [34]. When the pH of CR

solution increased from 3 to 6, the adsorption efficiency of AC to CR changed significantly. The adsorption capacity on CR was relatively best at $pH = 6$ (Fig. 4a), which may be caused by the formation of $-NH_3^+$ and $-SO_3H$ species at lower pH for the protonation of $-NH_2$ and $-SO_3^-$ functional groups in the CR molecule at lower pH [35]. The results

Table 2

Isotherm parameters for CR adsorption onto AC.

Adsorption isotherm	Fitting parameters	298 K	308 K	318 K
Langmuir	$q_{max}(mg/g)$	1544.53	1775.63	1960.99
	$K_L(L/mg)$	0.1647	0.2098	2.6619
	R^2_{adj}	0.9795	0.9816	0.9695
Freundlich	$K_F(L/mg)$	956.647	1060.626	1570.689
	$1/n$	0.088	0.099	0.049
	R^2_{adj}	0.8477	0.9449	0.8744
Liu	$q_{max}(mg/g)$	1509.60	1873.55	2006.04
	$K_g(L/mg)$	0.1245	0.2799	4.7952
	n_L	1.342	0.671	0.6271
	R^2_{adj}	0.9872	0.9957	0.9949

Table 3

Thermodynamic parameters for the adsorption of CR onto AC.

Temp. (K)	K_{eq}	$\ln K_{eq}$	ΔG^0 (kJ·mol ⁻¹)	ΔS^0 (J·K ⁻¹ ·mol ⁻¹)	ΔH^0 (kJ·mol ⁻¹)
298	8.67×10^4	11.371	-28.17	0.5775	144.84
308	1.95×10^5	12.181	-31.19		
318	3.34×10^5	15.022	-39.72		

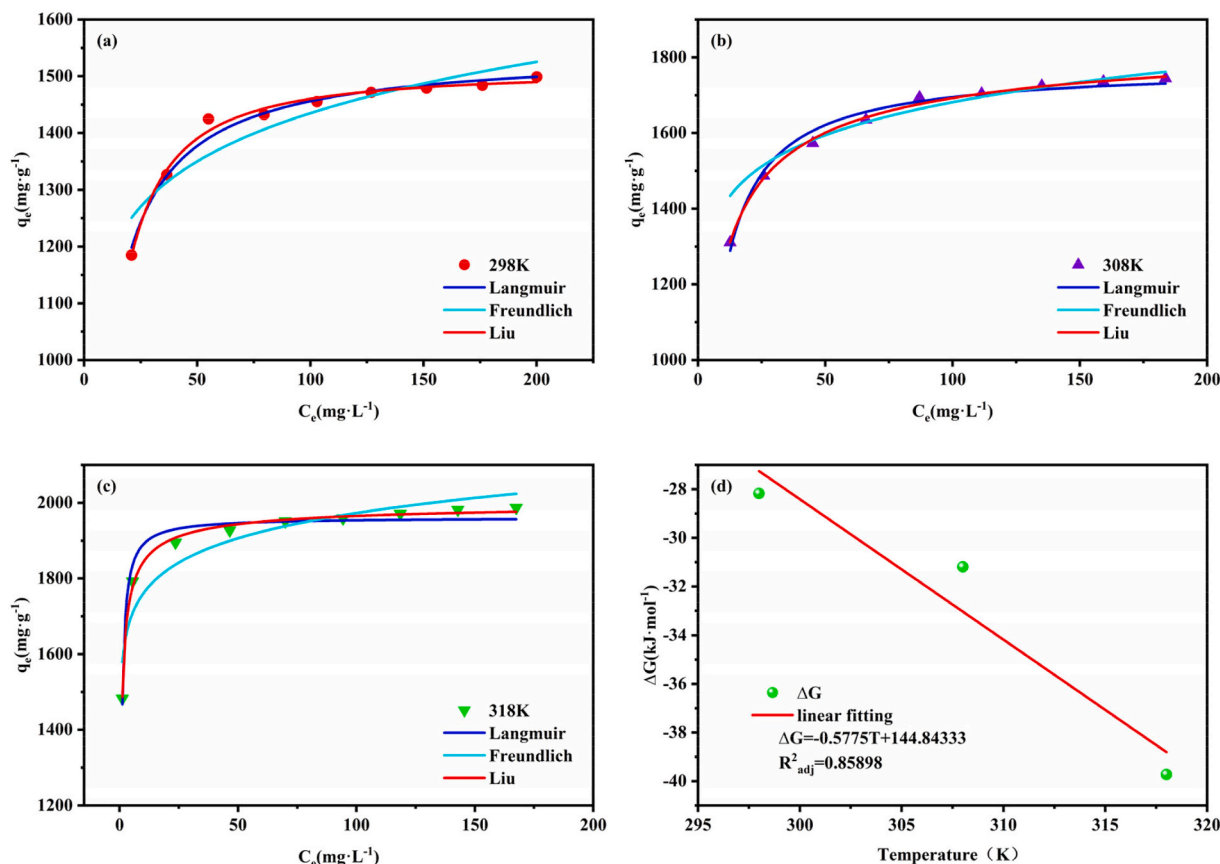


Fig. 5. (a,b and c) Fitting of adsorption isotherms at three different temperatures; (d) Linear fitting plots between ΔG^0 and K.

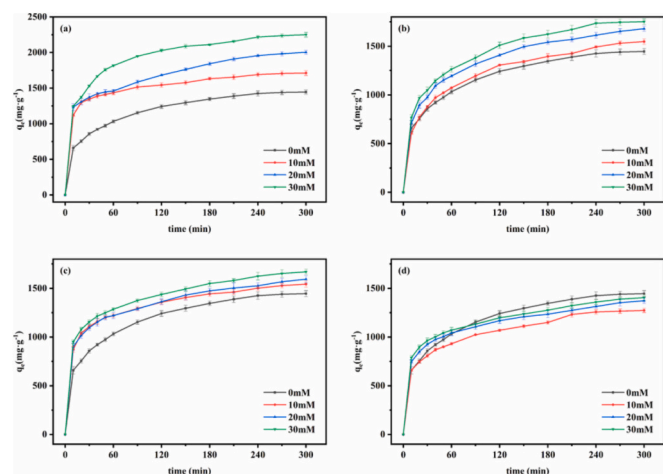


Fig. 6. CR adsorption efficiencies by AC in the presence of Ca^{2+} (a); and Na^+ (b); and K^+ (c); and NH_4^+ (d).

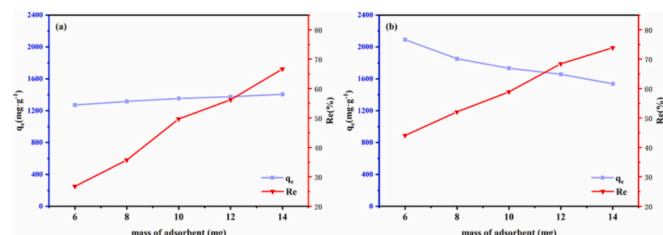


Fig. 7. Effect of adsorbent dosage on CR removal in the presence of without cation(a) and 10 mM Ca^{2+} (b).

showed that the positively charged AC repelled with the cationic functional groups on the CR surface, resulting in a decrease in the adsorption effect. On the other hand, the large amount of H^+ in the solution will compete with Congo red for the adsorption sites on AC, which will also lead to poorer adsorption of AC at low pH. As the pH of the solution increased, the positive charge on the surface of AC gradually decreased and the repulsive force with CR weakened. When $\text{pH} > 4$, the surface of AC is negatively charged, and the electrostatic mutual attraction with the positively charged functional group on top of Congo red is enhanced, and the adsorption effect gradually increases. The highest adsorption effect of AC on CR was achieved at pH 6.0 of the solution, when there was a good electrostatic interaction between AC and CR or van der Waals interaction between CR and AC molecules. When the pH gradually changed to basic, the functional groups on the surface of CR did not protonate. On the one hand, the negatively charged AC will strongly repel with the anion group on CR. On the other hand, as the amount of OH^- in the solution increases, it will compete with the CR for the adsorption sites on the AC, leading to a deterioration of the adsorption effect one by one [36]. Therefore, strong acid or base conditions are not favorable for the adsorption of Congo red dye.

3.2.2. Equilibrium of adsorption

As shown in Fig. 5 and Table 2, to investigate the adsorption mechanism, the classical Langmuir model, Freundlich model and Liu model were fitted to the experimental data. From the obtained fitting results, it can be seen that the correlation coefficient of Liu model can be as high as >0.98 with adsorption experiments under all of the temperature. As a result, the results show that the Liu model can better describe the kinetic behavior of CR adsorption on AC compared with other models, indicating that the adsorption of CR on AC is non-homogeneous, with different active sites acting simultaneously and with different free adsorption energies [37]. According to Table 2, the maximum

adsorption of AC on CR (q_{max}) increases significantly as the temperature increases from 298 K to 318 K, indicating that the adsorption of CR on AC is a heat absorption process. The high temperature favors the migration of CR molecules and facilitates the adsorption. The predicted maximum adsorption of CR on activated carbon samples obtained by fitting the Liu model was 1509.60 mg/g, and this result was also very close to the experimental value.

3.2.3. Adsorption thermodynamic evaluation

The adsorption data of AC on CR at different temperatures were used to calculate the relevant thermodynamic parameters required for the experimental analysis based on the thermodynamic equations of Table S1. As shown in Fig. 5d and Table 3, ΔG^0 was negative at three different temperatures and its absolute value increased with increasing temperature, indicating that the adsorption reaction of CR on AC was spontaneous [38]. The calculated positive value of ΔH^0 indicates that the adsorption reaction of AC on CR is heat-absorbing and the high temperature is favorable for the adsorption reaction, which is consistent with the adsorption isotherm analysis above. The positive value of ΔS^0 can indicate the stochastic nature of the adsorption process [39].

3.3. Effect of cation on CR adsorption

Cations in solution can affect the adsorption capacity of AC [40]. Due to the complexity of the environmental water composition, a wide variety of ions are presented in wastewater systems. The cations present in the wastewater will compete with the CR for the active adsorption sites on the AC [41,42]. Therefore, the four most common ions (Ca^{2+} , Na^+ , K^+ , NH_4^+) in dye wastewater were selected to study the effects of cation on CR adsorption. The concentrations of the four cations were 10 mM, 20 mM and 30 mM, respectively. The experiments were carried out separately using different cations of chloride salts to minimize the error. Different concentrations of NH_4Cl , NaCl , KCl , CaCl_2 were added to 200 mg/L of CR dye solution.

As shown in Fig. 6a, Ca^{2+} in solution can substantially enhance the adsorption effect of AC on CR, and the adsorption of AC on CR could reach 1712 mg/g at a Ca^{2+} concentration of 10 mM, which increased the adsorption effect by 18.48 % compared with that of no cations. This may be due to the fact that CR is an anionic dye with negative charge, and the presence of large amount of Ca^{2+} can make the activated carbon positive charged, thus enhancing the adsorption effect of adsorbent on CR. As the concentration of Ca^{2+} increased, the adsorption rate and maximum adsorption capacity of CR continued to increase. When the concentration of Ca^{2+} increased to 20 mM and 30 mM, the maximum adsorption of AC on CR increased by 37.62 % and 54.65 %, respectively, and could reach a maximum of 2250.69 mg/g, which was much higher than the adsorption effect of AC without the addition of cations.

The presence of Na^+ and K^+ also enhanced the adsorption effect of AC on CR. As shown in Fig. 6b and Fig. 6c, when the concentration of Na^+ and K^+ were 10 mM, the adsorption effect of AC on CR was 1547.41 and 1543.51 mg/g. when the concentration of Na^+ and K^+ in the solution turned to be 20 mM and 30 mM, the adsorption capacity of AC to CR has increased by 15.39 % & 20.36 % and 9.53 % & 14.72 % respectively. Experiments showed that the presence of Na^+ and K^+ could enhance the adsorption effect of AC on CR to some extent.

As shown in Fig. 6d, the presence of NH_4^+ ions did not enhance the adsorption effect of AC on CR. When the concentration of NH_4^+ ions in the solution was 10 mM, the adsorption effect of AC on CR was 1273.06 mg/g, which was 12.52 % lower than when no cation was added. This may be attributed to the fact that NH_4^+ occupies the adsorption sites on the activated carbon surface and inhibits the adsorption of CR. It is noteworthy that NH_4^+ promotes AC less than alkali metal cations [43]. However, as the concentration of NH_4^+ ions continued to increase, the adsorption effect of AC on CR also increased, indicating that the large amount of NH_4^+ ions made the surface of AC positively charged, which increased the adsorption effect to some extent.

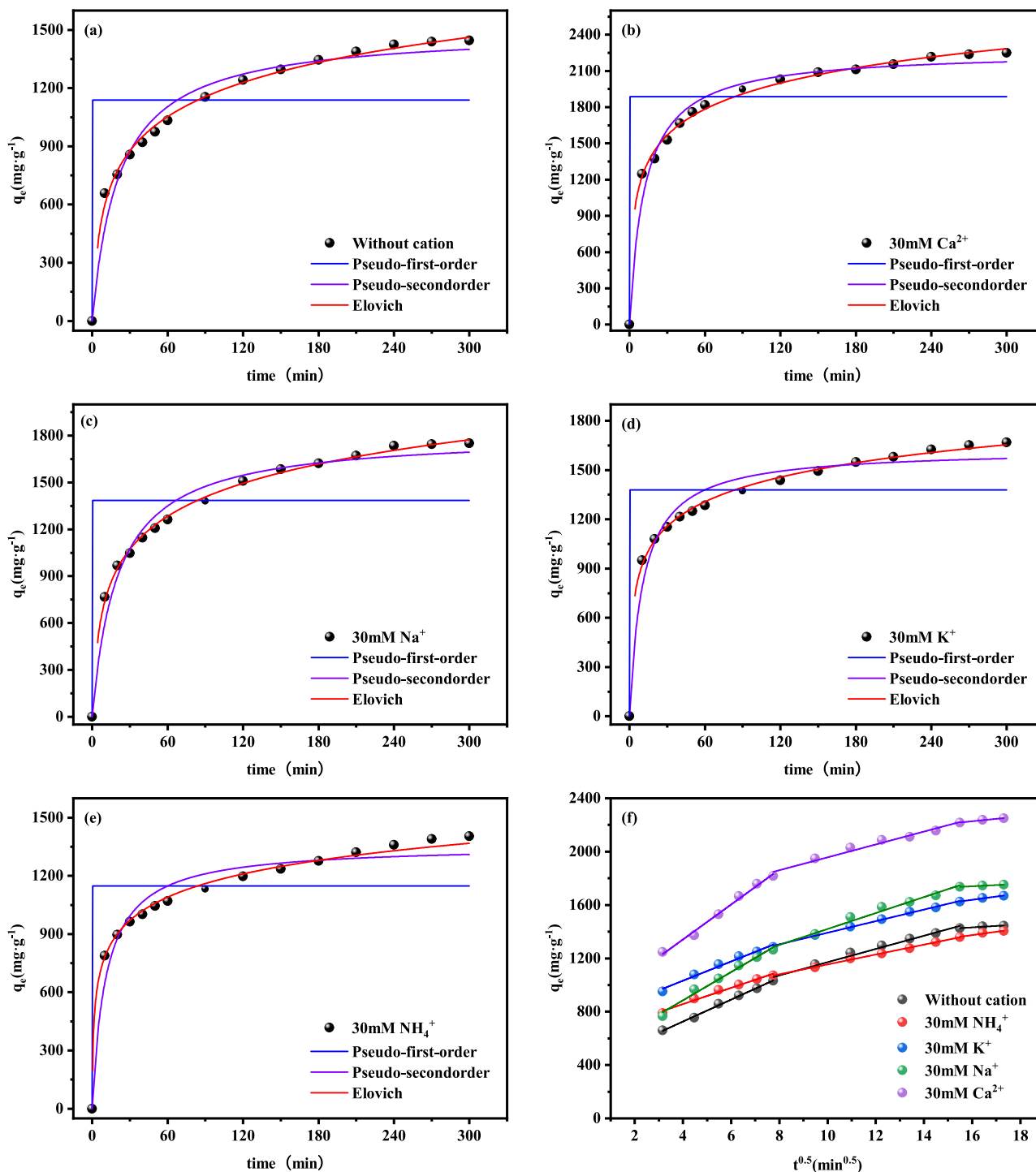


Fig. 8. Adsorption kinetics of CR on AC in the presence of without cation(a); Ca^{2+} (b); and Na^{+} (c); and K^{+} (d); and NH_4^{+} (e). Adsorption kinetics of CR on AC by intra-particle diffusion model(f).

It can be concluded that Ca^{2+} , Na^{+} , and K^{+} can promote the adsorption of CR by AC. (i) Change of hydrophobicity, inorganic salt cations such as Ca^{2+} , K^{+} , Na^{+} and NH_4^{+} affect the dispersion of dyes in aqueous solution and change the interaction between AC and CR. Specifically, the addition of electrolytes in CR solutions competes more strongly for solvent molecules than CR dyes, leading to the migration of solvent molecules from CR dyes to cationic compounds, which reduces the solubility of CR and makes it more amenable to adsorption by AC. Moreover, the effect of Ca^{2+} is stronger than Na^{+} and K^{+} , which leads to a higher adsorption capacity of AC on CR [44,45]. (ii) Adsorption

molecular complexation. The cation will complex with the organic dye molecule to form a surface complex with more affinity for the AC adsorption site [46]. The formation of this new complex will promote the adsorption of AC. (iii) Creating bonds with metal. The cations may act as a bridge between AC and CR, resulting in an AC- Ca^{2+} -CR-like structure [47]. This new structure will also facilitate the adsorption. (iv) Electrostatic effect. The adsorption experiments were carried out at pH 7. The zeta potential analysis of AC in Fig. 4a shows that the surface of AC is negatively charged at this time. AC generates a large electrostatic repulsion with the anionic dye CR, which reduces the adsorption effect

Table 4Adsorption kinetic parameters for CR on different cation types. $\alpha(\text{mg} \cdot (\text{g} \cdot \text{min})^{-1})$.

Kinetics equation	Parameters	Without cation	30 mM Ca^{2+}	30 mM Na^+	30 mM K^+	30 mM NH_4^+
Pseudo-first-order	$Q_e (\text{mg} \cdot \text{g}^{-1})$	1138.169	1888.165	1358.674	1379.596	1148.851
	$k_1 (\text{min}^{-1})$	212.515	351.193	258.941	257.825	214.402
	Adj. R^2	0.560	0.700	0.569	0.7228	0.715
Pseudo-second-order	$Q_e (\text{mg} \cdot \text{g}^{-1})$	1500.435	2262.927	1809.654	1626.638	1358.701
	$k_2 (\text{g} \cdot (\text{mg} \cdot \text{min})^{-1})$	3.096×10^{-5}	3.651×10^{-5}	2.703×10^{-5}	5.722×10^{-5}	6.689×10^{-5}
	Adj. R^2	0.972	0.981	0.976	0.965	0.961
Elovich	$\alpha(\text{mg} \cdot (\text{g} \cdot \text{min})^{-1})$	266.518	1602.410	343.911	297.223	1175.672
	$\beta (\text{g} \cdot \text{mg}^{-1})$	0.004	0.003	0.003	0.005	0.005
	Adj. R^2	0.997	0.997	0.998	0.998	0.997
Intra-particle diffusion	$K_1(\text{mg}(\text{g} \cdot \text{min}^{0.5})^{-1})$	82.234	60.826	106.088	72.123	131.205
	Adj. R^2	0.997	0.979	0.980	0.977	0.989
	$K_2(\text{mg}(\text{g} \cdot \text{min}^{0.5})^{-1})$	49.572	37.038	59.735	43.402	48.263
	Adj. R^2	0.980	0.998	0.979	0.996	0.969
	$K_3(\text{mg}(\text{g} \cdot \text{min}^{0.5})^{-1})$	10.992	24.944	8.906	24.365	18.064
	Adj. R^2	0.920	0.929	0.975	0.976	0.979

Table 5Adsorption capacities (q_{max}) of CR on various adsorbents.

Adsorbents	q_{max} (mg/g)	Ref.
Edamame shell based activated carbon	1509.60	This work
Edamame shell based AC in the presence of Ca^{2+}	2250.69	This work
Casuarina empty fruit-based activated carbon	232.00	[17]
kenaf-based activated carbon	14.20	[56]
Commercial activated carbon	300.00	[57]
Bombax Buonopozense bark activated-carbon	510.20	[58]
ZnFe_2O_4 : 1 wt% Ce^{3+} /Carbon fibers composite	265.20	[59]
Apricot stone	32.85	[60]
Guar gum/activated carbon nanocomposite	831.82	[61]
Activated carbon with electropositive amine modification	319.39	[62]

of AC. With the increase of cation concentration in the CR solution, more cations would surround the hydroxyl groups with negative charge on the AC surface. As a result, the electrostatic repulsion between the negatively charged AC surface and the negatively charged CR anion is weakened, which is favorable for CR adsorption [48]. Higher concentrations of cations also positively charge the surface of AC, thus increasing the electrostatic interaction between AC and anionic dyes. All these enhance the adsorption effect of AC. The enhancement of CR adsorption by AC by Ca^{2+} is most pronounced because Ca^{2+} carries more positive charge and can make AC more positively charged [49].

3.4. Effect of adsorbent mass

By maintaining the initial concentration of CR at a high 200 mg/L, the effect of adsorbent dosage on CR adsorption in the presence or absence of cations in the solution was analyzed by varying the amount of adsorbent added. The cation chosen for the experiment was Ca^{2+} with an ion concentration of 10 mM. As can be seen from the Fig. 7a, when the mass of AC was increased from 6 mg to 14 mg, the CR removal rate increased from 26.81 % to 66.70 %. The improvement of CR removal by increasing the amount of adsorbent is related to the increase of the number of AC active sites [50,51]. The unit adsorption capacity of AC did not show a significant decrease, probably due to the low repulsive force between the adsorbents. As shown in Fig. 7b, at a Ca^{2+} concentration of 10 in the solution, the unit adsorption appears opposite to the removal rate. This means that during the first stage of the adsorption process, there are a large number of free adsorption sites available to adsorb CR, and then over time, there is competition between adsorbents for Ca^{2+} . The decrease in adsorption at this stage is attributed to the decrease in Ca^{2+} that can be bound per unit adsorbent and the decrease in Ca^{2+} contribution to CR per unit adsorbent. Also probably due to the

repulsive forces between adsorbents, fewer free adsorption sites remain, leading to a decrease in the adsorption effect of CR [52,53].

3.5. Kinetics analysis

In order to investigate the kinetics of CR adsorption by activated carbon and explore the adsorption mechanism, each set of experimental data at a cation concentration of 30 mM was fitted using the pseudo-first order, pseudo-second order, Elovich model, and intraparticle diffusion equations. The adsorption experimental data and kinetic curves are shown in Fig. 8 and the fitting results are presented in Table 4. It is not difficult to find that the addition of cation did not change the kinetic mode of AC to CR adsorption. For the presence of different ions, the quasi-second-order model fits better than the quasi-first-order model for the adsorption process of CR in the presence of Ca^{2+} , Na^+ , K^+ , NH_4^+ . Kinetic model R^2 were 0.981, 0.976, 0.965 and 0.961, respectively. From the above results, it can be found that in the presence of various cations, the adsorption process of AC on CR is not only a simple physical adsorption, but a complex adsorption process in which physical adsorption and chemical adsorption co-exist. And the two adsorptions have synergistic effects in the whole adsorption process of AC [54].

Using Elovich model, the linear correlation coefficients of the adsorption model without the addition of ions were 0.997, and after the addition of ions the linear correlation coefficients were 0.997, 0.998, 0.998, 0.997, which showed good linear correlation, indicating that the adsorption process of CR adsorbate interacts with different types of adsorption chemical centers and belongs to non-homogeneous adsorption. As shown in Fig. 8f, the diffusion process of CR on AC in the presence of different cations was analyzed using an intraparticle diffusion model of CR on AC. The adsorption process can be divided into three stages: The first stage is the film diffusion process of CR from the solution to the outer surface of AC, and the adsorption rate of AC in the presence of Ca^{2+} in this process is much higher than that of other cations as well as in the absence of ions. The second stage is the rate-limiting stage, where the adsorption rate of AC gradually decreases due to the reduction of active sites available for adsorption on the AC surface. The third stage is the adsorption equilibrium stage, when the adsorption of CR by AC reaches saturation. However, since the second segment of the line obtained from the model does not pass through the coordinates origin, the adsorption process is not completely controlled by intraparticle diffusion [55]. According to the results in Table 4, metal cations significantly increased the adsorption rate and capacity of AC on CR.

3.6. Comparison of different adsorbents

Table 5 compares the maximum adsorption capacity (q_{max}) of AC with the biosorbents used by other researchers for CR. From the table, the AC prepared in this paper has better adsorption performance

compared with other biosorbents. Generally, it is difficult to enhance the adsorption effect of the same activated carbon on CR. From Table 4, it can be seen that the adsorption effect of AC on CR obtained a great enhancement in the presence of Ca^{2+} , so the efficient adsorption of CR can be achieved by appropriately increasing the concentration of Ca^{2+} in the CR solution. Therefore, it can be inferred from Table 5 that AC adsorbent can adsorb and recover CR from wastewater very efficiently and economically, and it is abundant and low cost, which has good application prospects.

4. Conclusions

In summary, an activated carbon with high specific surface area ($2653.382 \text{ m}^2\text{g}^{-1}$) and pore volume ($1.324 \text{ cm}^3\text{g}^{-1}$) was successfully prepared from edamame shells, which had good adsorption performance on CR. The addition of alkali metal cations enhanced the adsorption of AC on CR, especially the addition of Ca^{2+} greatly improved the adsorption effect of AC on CR, and its adsorption amount could reach $2250.69 \text{ mg}\cdot\text{g}^{-1}$. The adsorption of CR on activated carbon is a spontaneous heat absorption process. Compared with most other adsorbents reported in the literature, the AC in this study showed better adsorption performance on CR. Studies investigating the addition of cations to enhance the adsorption capacity of adsorbents for anionic dyes are also of great practical application. Therefore, it can be well applied to agricultural waste treatment and dye wastewater treatment, and has good prospects for application in the field of environmental protection.

CRedit authorship contribution statement

Song He: Conceptualization, Methodology, Writing – review & editing. **Junwei Sun:** Writing – original draft. **Qilin Chen:** Visualization. **Xin Jin:** Visualization. **Xiya Wu:** Validation. **Fuliang Tian:** Validation. **Xiaoqian Zhang:** Validation. **Ping Li:** review & editing. **Hongliang Sheng:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgements

This research was financially supported by the Natural Science Foundation of China (No. 52074201 & 51706165).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.diamond.2023.109930>.

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